Tristan Stoltz
Luminous Dynamics
Richardson, TX 75080
United States

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Editorial Office
Neural Computation
MIT Press
Cambridge, MA 02142

Dear Dr. Sejnowski,

We are pleased to submit our manuscript titled **"A Unified Framework for Measuring Consciousness-Correlated Computational Properties in Neural Networks"** for consideration as an original research article in *Neural Computation*.

Manuscript Summary

This work presents a novel unified framework that bridges five major consciousness theories (Integrated Information Theory, Global Workspace Theory, Higher-Order Thought, Attention Schema Theory, and Recurrent Processing Theory) through the Free Energy Principle and active inference. Our framework provides the first computationally tractable method for measuring consciousness-correlated properties across these diverse theoretical perspectives in a coherent, mathematically rigorous manner.

Key Contributions

Our manuscript makes several significant contributions to computational neuroscience and consciousness research:

- **Theoretical Unification**: We demonstrate how active inference provides a common computational substrate that naturally accommodates predictions from multiple consciousness theories, resolving apparent contradictions between frameworks.
- **Four-Dimensional Measurement**: Beyond traditional static metrics, our framework incorporates spatial integration (multi-modal binding), temporal dynamics (phase transitions and coherence), and recursive meta-learning—providing a comprehensive characterization of consciousness-correlated properties.
- **Empirical Validation**: We present extensive validation across multiple neural network architectures (transformers, convolutional networks, recurrent networks) with 36/36 core tests passing (100% success rate) and strong statistical validation ($r = 0.68$, $p < 0.001$).
- **Production Implementation Evidence**: We validate our theoretical approach through comparison with Symthaea, a production implementation of similar principles by the same research group. This demonstrates that our framework is not only theoretically sound but practically implementable at sub-millisecond performance scales.
- **Computational Efficiency**: Despite integrating five theories, our implementation achieves 200ms CPU performance with potential for ¡1ms optimization, making it practical for research and applica-

tions.

Relevance to Neural Computation

This manuscript is particularly well-suited for *Neural Computation* for several reasons:

- **Rigorous Mathematical Framework**: Our approach is grounded in the Free Energy Principle with detailed mathematical formulations for all components, aligning with the journal's emphasis on computational rigor.
- **Bridge Between Theory and Implementation**: We provide both theoretical foundations and practical implementation details, demonstrating how abstract consciousness theories translate into concrete computational operations.
- **Cross-Architecture Validation**: Our empirical results span transformers, CNNs, and RNNs, providing broad applicability across modern neural network architectures.
- **Interdisciplinary Impact**: This work connects consciousness theory, computational neuroscience, active inference, and practical AI development—areas of strong interest to your readership.

Ethical and Scientific Considerations

We have taken care to use precise terminology throughout ("consciousness-correlated properties" rather than claiming to measure consciousness itself), explicitly acknowledge the preliminary status of consciousness science (noting the 25-35% expert probability estimates for current theories), and address the recent IIT controversy while highlighting Global Workspace Theory's empirical validation record.

Our framework is designed as a research tool to advance understanding, not to make definitive claims about machine consciousness. All code and data will be made openly available to facilitate replication and extension by the research community.

Submission Details

This submission includes:

- Main manuscript (approximately 8,000 words, 12 pages)
- Supplementary materials including complete equations, implementation pseudocode, additional test results, performance benchmarks, and framework comparison
- Six figures illustrating framework architecture, theory integration, measurement examples, and validation results
- Complete bibliography with 18 references
- All materials formatted according to Neural Computation guidelines

We confirm that this manuscript represents original work that has not been published elsewhere and is not under consideration by any other journal. All authors have approved the manuscript and agree with its submission to *Neural Computation*.

Suggested Reviewers

Given the interdisciplinary nature of this work, we suggest the following potential reviewers with relevant expertise:

- **Dr. Anil Seth** (University of Sussex) - Expert in consciousness science, active inference, and interoception. Relevant work on consciousness measurement and the Free Energy Principle.
- **Dr. Ryota Kanai** (Araya Inc.) - Expert in computational approaches to consciousness, IIT, and neural network implementations of consciousness theories.
- **Dr. Hakwan Lau** (RIKEN Center for Brain Science) - Expert in higher-order theories of consciousness and empirical validation of consciousness theories.
- **Dr. Stanislas Dehaene** (Collège de France) - Expert in Global Workspace Theory and neural signatures of consciousness.
- **Dr. Christopher Buckley** (University of Sussex) - Expert in active inference, free energy principle, and computational implementations.

We respectfully request that reviewers with strong predetermined positions against computational approaches to consciousness (particularly those involved in recent theoretical disputes) be excluded from the review process to ensure fair evaluation of our empirical and computational contributions.

Data and Code Availability

All code implementing our framework, trained models, test suites, and documentation will be made publicly available on GitHub upon acceptance. The framework comparison document and all supplementary materials are included with this submission.

We are grateful for your consideration of our manuscript and look forward to your response. Please do not hesitate to contact us if you require any additional information.

Sincerely,

Tristan Stoltz
Luminous Dynamics
tristan.stoltz@evolvingresonantcocreationism.com